

Desmos Exploration: Relations of the Form $xy = k$

Learning goal: I can use technology to explore and explain properties of relations of the form $xy = k$ and relate them to algebraic and graphical representations.

Review

Navigate to <https://desmos.com>, and then answer the following questions:

1. Graph the equation $y=3x+7$. On a new line, type $y=mx+3$, then click “add slider.”
 - Adjust the slider to find the value of the slope that makes the line perpendicular to the original.
 - Does this match what we learned about perpendicular lines?
2. Graph the following 4 equations: $y=2$, $y=-2$, $x=2$, and $x=-2$. Then, modify them so that they form a square with area 36.
3. Graph a non-linear equation using Desmos, then reproduce it below:

Main Lesson - Relations of the form $xy = k$

Previously, we have been discussing relations of the form $y = mx + b$, which we called *linear relations*. We called them this because they take the form of a line when we plot them on a graph. We also discussed the existence of *non-linear relations*, but we didn't really elaborate on them.

Today, we will focus on a particular category of non-linear relation which is referred to as an **inverse proportionality** relation.

Getting Started

Using Desmos, plot $y = \frac{1}{x}$ and answer the following questions:

1. As x increases, what happens to y ?
2. Identify and plot at least three points on the graph
3. What happens to y as x gets close to 0?

Inverse proportionality comes up very commonly in Math and Science. Consider the example of Ohm's Law:

$$V = IR \leftrightarrow I = \frac{V}{R}$$

The second version of the equation expresses an inverse proportionality relation between current (I) and resistance (R). That is, for a fixed value of voltage (V), increasing resistance (R) causes a decrease in the current (I).

In contrast, voltage and current are **proportional**. That is, an increase in current causes an increase in voltage.

Back to Desmos

- In your first line, set voltage to be a specific value, e.g. 9V. In Desmos, this will look like $V=9$.
- Next, add in your expression for Ohm's Law: $I=\frac{V}{R}$

Use your curve to fill in the following table:

Voltage (V)	Resistance (Ω)	Current (A)
9	10	
9	100	
9		9
9		90

Now, set your voltage to a different number and create a similar table with at least 4 rows below.

What effect does changing the voltage have?

Inverse Proportionality

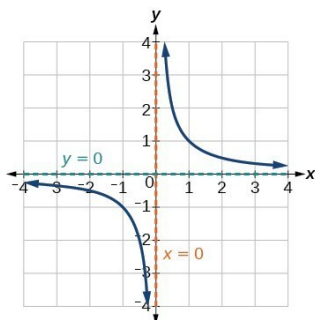
We say that two quantities x and y are inversely proportional if they're related by the expression

$$y = \frac{k}{x}$$

where k is just a number to scale the relationship, similar to m in linear relations.

We commonly write these relations as $xy = k$ instead. For example, $y = \frac{3}{x}$ can be rewritten as $xy = 3$. The benefit of writing it this way is that it makes it so that the expression on the left is not undefined anywhere (namely, at 0).

Unless $k = 0$, you can never have that $x = 0$ or $y = 0$. As x gets closer and closer to 0, the value of y *diverges* to positive or negative infinity.



Investigate this using Desmos. The closer you get to zero from the right, the larger the number gets. In math, we call a line that you approach but never cross an **asymptote**.

Determining value of k

Unlike with linear relations, we only need to know one point to figure out the value of k .

Example Using Desmos,

- Plot the point $(3, 7)$
- Write $xy=k$, then click “add slider”
- Determine a value for k that makes the line pass through the point

Alternatively, you can determine the value of k using algebra. Simply plug in the known point to the equation, then solve for k .

Example 2 Using algebra or Desmos, find the value of k such that $xy = k$ passes through $(100, 19)$.

Inequalities

Use Desmos to sketch $xy < 4$. Describe what you see. Why does it look this way?

Without graphing immediately, guess what $xy \geq 3$ will look like:

Practice Questions

1. Draw graphs of the following relations. You may use Desmos to help determine the shape.

a) $xy = 2$

b) $xy = -3$

c) $xy = \frac{1}{2}$

d) $xy = -\frac{3}{4}$

2. What is the difference between $xy = k$ and $y = \frac{k}{x}$?

3. In the following relations, what happens to y as x increases? Identify any asymptotes.

a) $xy = 4$

b) $y = 3x + 4$

c) $y = -\frac{3}{x}$

4. Consider the relation given by the equation $y = \frac{4}{x} + 3$. Graph the equation in Desmos. What does the $+3$ represent geometrically? What is the equation of the horizontal asymptote?